INCH-POUND
MIL-M-38510/308A
AMENDMENT 3
28 November 1997
SUPERSEDING
AMENDMENT 2
13 September 1985

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL,
ARITHMETIC LOGIC UNITS/FUNCTION GENERATORS, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER 6 SEPTEMBER 1996

This amendment forms a part of MIL-M-38510/308A, dated 2 February 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute title as printed on this amendment.

1.1, line two: Delete "Three" and substitute "Two".

1.2.3, case outline delete the following: "L F-7 (24-lead, 3/8" x 1/2", flat package)".

Add the following: "3 C-4 (28-terminal, .450" x .450" x .450"), square chip carrier package".

1.3, thermal resistance: Delete "0.04 °C/mW for flat package;
0.02 °C/mW for dual-in-line package"

and substitute "Case 3 - - - - - - - - - - - - - - - 0.08°C/mW 2/
Cases J, K, Z - - - - - - - - - - - (See MIL-M-38510, appendix C)"

1.3, Junction temperature, after (T_J) add "3/".

Add at bottom of page:

"2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the values stated herein.

"3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions, as per method 5004 of MIL-STD-883."

PAGE 2

3.2.5, Schematic circuits, delete and substitute:

"3.2.5 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

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* The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
13A	New page
13B	New page
13C	New page

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PAGE 3

TABLE I, conditions column heading: Add "-55 ˚C ≤ T_C ≤ +125˚C". Make this same change on pages 4 through 6.

PAGE 6

TABLE I, conditions column heading: Add "1/".

PAGE 7

3.5, delete and substitute:

"3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

3.6, delete the second sentence.

TABLE II, test requirement: Delete "Interim" and substitute "Initial".

TABLE II, group B test requirements: Add "Subgroup 5".

TABLE II: Delete class C device column in its entirety.

4.2b Delete "Interim" and substitute "Initial" (2 places).

PAGE 8

4.2c, delete and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

4.4.3, delete and substitute:

"4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

(1) Test condition D or E, using the circuit shown on figure 5, or equivalent.

(2) T_A = +125˚C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

4.6, delete in its entirety.

PAGE 9

FIGURE 1: Delete "Cases J, K, L, and Z" and substitute "Cases J, K, Z". Add new figure for case 3 as printed on page 3 of this amendment.

Pin 1: Delete "B_0" and substitute "$\overline{B}_0$".
Pin 2: Delete "A_0" and substitute "$\overline{A}_0$".
Pin 9: Delete "F_0" and substitute "$\overline{F}_0$".
Pin 10: Delete "F_1" and substitute "$\overline{F}_1$".
Pin 11: Delete "F_2" and substitute "$\overline{F}_2$".
Pin 17: Delete "$\overline{6}$ " and substitute "$\overline{G}$".

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DEVICE TYPE 01

CASE 3

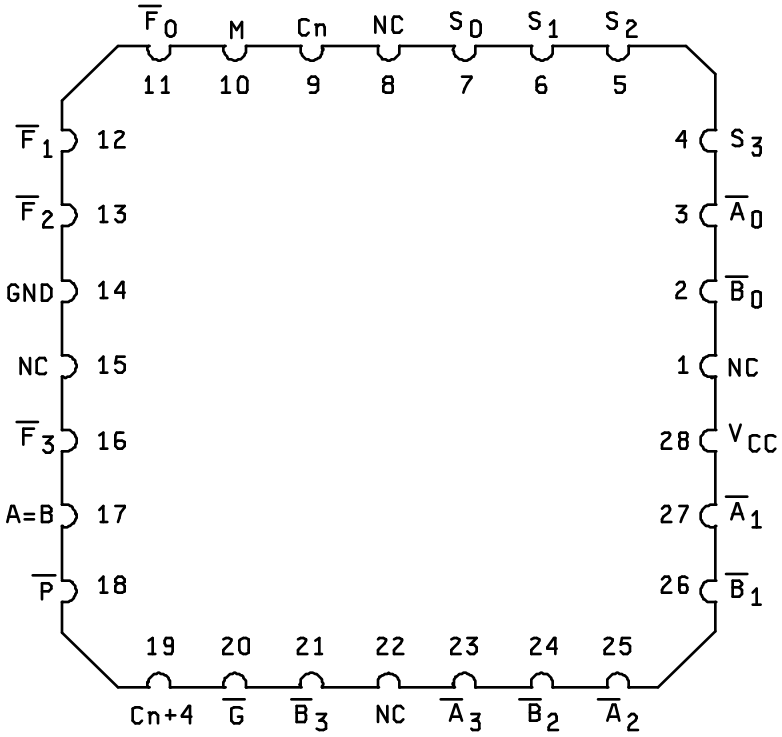

FIGURE 1. Terminal connections (top view).

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PAGE 15

TABLE III, device type 01, under cases, delete pin designations and substitute the following: Make these same changes on pages 16 through 20.

"

Cases	Pins not referenced are NC											
J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12
3	2	3	4	5	6	7	9	10	11	12	13	14
Test no.	$\overline{B}_0$	$\overline{A}_0$	S_3	S_2	S_1	S_0	C_n	M	$\overline{F}_0$	$\overline{F}_1$	$\overline{F}_2$	GND

Cases	Pins not referenced are NC											
J,K,Z	13	14	15	16	17	18	19	20	21	22	23	24
3	16	17	18	19	20	21	23	24	25	26	27	28
Test no.	$\overline{F}_3$	A=B	$\overline{P}$	C_n+4	$\overline{G}$	$\overline{B}_3$	$\overline{A}_3$	$\overline{B}_2$	$\overline{A}_2$	$\overline{B}_1$	$\overline{A}_1$	V_{CC}

PAGE 20

Footnote 4/: Delete "115" and substitute "120".

PAGE 21

6.1, Notes: Delete in its entirety.

6.2, delete and substitute:

"6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes."

6.4, delete "GND - - - - - - - - - Electrical ground (common terminal)". and substitute "GND - - - - - - Ground zero voltage potential".

Custodians:
Army - CR
Navy - EC
Air Force - 17

Preparing activity:
DLA-CC

Review activities:
Army - AR, MI, SM
Navy - OS, SH, AS, CG, MC
Air Force - 19, 85, 99

(Project 5962-1810)

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$\overline{SUM}$ MODE TEST TABLE

FUNCTION INPUTS: S0 = S3 = 4.5 V, S1 = S2 = M = 0V

TEST	INPUT UNDER TEST	OTHER INPUT SAME BIT		OTHER DATA INPUTS		OUTPUT UNDER TEST	OUTPUT WAVEFORM
		APPLY 4.5 V	APPLY GND	APPLY 4.5 V	APPLY GND		
t_{PLH1} / t_{PHL1}	$\overline{A}_i$ 1/	$\overline{B}_i$	None	Remaining $\overline{A}$ and $\overline{B}$	C_n	Any $\overline{F}_i$	1
t_{PLH1} / t_{PHL1}	$\overline{B}_i$	$\overline{A}_i$	None	Remaining $\overline{A}$ and $\overline{B}$	C_n	Any $\overline{F}_i$	1
t_{PLH3} / t_{PHL3}	$\overline{A}_i$	$\overline{B}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{P}$	1
t_{PLH3} / t_{PHL3}	$\overline{B}_i$	$\overline{A}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{P}$	1
t_{PLH5} / t_{PHL5}	$\overline{A}_i$	None	$\overline{B}_i$	Remaining $\overline{B}$	Remaining $\overline{A}$, C_n	$\overline{G}$	1
t_{PLH5} / t_{PHL5}	$\overline{B}_i$	None	$\overline{A}_i$	Remaining $\overline{B}$	Remaining $\overline{A}$, C_n	$\overline{G}$	1
t_{PLH7} / t_{PHL7}	C_n	None	None	All $\overline{A}$	All $\overline{B}$	Any $\overline{F}_i$	1
t_{PLH9} / t_{PHL9}	C_n	None	None	All $\overline{A}$	All $\overline{B}$	C_{n+4}	1
t_{PLH11} / t_{PHL11}	$\overline{A}_i$	None	$\overline{B}_i$	Remaining $\overline{B}$	Remaining $\overline{A}$, C_n	C_{n+4}	2
t_{PLH11} / t_{PHL11}	$\overline{B}_i$	None	$\overline{A}_i$	Remaining $\overline{B}$	Remaining $\overline{A}$, C_n	C_{n+4}	2

1/ In $\overline{A}_i$, $\overline{B}_i$ and $\overline{F}_i$, i= 0, 1, 2, or 3.

FIGURE 4. Waveforms for propagation delay time and test circuit for device type 01 - Continued.

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D TFF MODE TEST TABLE

FUNCTION INPUTS: S1 = S2 = 4.5 V, S0 = S3 = M = 0 V

TEST	INPUT UNDER TEST	OTHER INPUT SAME BIT		OTHER DATA INPUTS		OUTPUT UNDER TEST	OUTPUT WAVEFORM
		APPLY 4.5 V	APPLY GND	APPLY 4.5 V	APPLY GND		
t_{PLH2} / t_{PHL2}	$\overline{A}_i$ 1/	None	$\overline{B}_i$	Remaining $\overline{A}$	Remaining $\overline{B}$, C_n	Any $\overline{F}_i$	1
t_{PLH2} / t_{PHL2}	$\overline{B}_i$	$\overline{A}_i$	None	Remaining $\overline{A}$	Remaining $\overline{B}$, C_n	Any $\overline{F}_i$	2
t_{PLH4} / t_{PHL4}	$\overline{A}_i$	None	$\overline{B}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{P}$	1
t_{PLH4} / t_{PHL4}	$\overline{B}_i$	$\overline{A}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{P}$	2
t_{PLH6} / t_{PHL6}	$\overline{A}_i$	$\overline{B}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{G}$	1
t_{PLH6} / t_{PHL6}	$\overline{B}_i$	None	$\overline{A}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{G}$	2
t_{PLH8} / t_{PHL8}	$\overline{A}_i$	None	$\overline{B}_i$	Remaining $\overline{A}$	Remaining $\overline{B}$, C_n	A = B	1
t_{PLH8} / t_{PHL8}	$\overline{B}_i$	$\overline{A}_i$	None	Remaining $\overline{A}$	Remaining $\overline{B}$, C_n	A = B	2
t_{PLH10} / t_{PHL10}	C_n	None	None	All $\overline{A}$ and $\overline{B}$	None	C_{n+4}	1
t_{PLH12} / t_{PHL12}	$\overline{A}_i$	$\overline{B}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	C_{n+4}	2
t_{PLH12} / t_{PHL12}	$\overline{B}_i$	None	$\overline{A}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	C_{n+4}	1

1/ In $\overline{A}_i$, $\overline{B}_i$ and $\overline{F}_i$, i= 0, 1, 2, or 3.

FIGURE 4. Waveforms for propagation delay time and test circuit for device type 01 - Continued.

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LOGIC MODE TEST TABLE

FUNCTION INPUTS: S1 = S2 = M = 4.5 V, S0 = S3 = M = 0 V

TEST	INPUT UNDER TEST	OTHER INPUT SAME BIT		OTHER DATA INPUTS		OUTPUT UNDER TEST	OUTPUT WAVEFORM
		APPLY 4.5 V	APPLY GND	APPLY 4.5 V	APPLY GND		
t_{PLH13}	$\overline{A}_i$ 1/	None	$\overline{B}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	Any $\overline{F}_i$	1
t_{PHL13}							
t_{PLH13}	$\overline{B}_i$	None	$\overline{A}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	Any $\overline{F}_i$	2
t_{PHL13}							

1/ In $\overline{A}_i$, $\overline{B}_i$ and $\overline{F}_i$, i= 0, 1, 2, or 3.

FIGURE 4. Waveforms for propagation delay time and test circuit for device type 01 - Continued.

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